

1. BFS — Breadth First Search:

Breadth First Search (BFS) is a graph traversal algorithm that visits the vertices level by level. It uses a queue to store the vertices that need to be visited.

ALGORITHM:

1. Start from the given vertex.
2. Mark the starting vertex as visited.
3. Insert the starting vertex into the queue.
4. Remove a vertex from the front of the queue.
5. Visit all its unvisited adjacent vertices.
6. Mark each visited vertex and insert it into the queue.
7. Repeat until the queue becomes empty.
8. Display the BFS traversal

Time Complexity:

Case	Time Complexity
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Best Case	$O(V + E)$
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Average Case	$O(V + E)$
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Worst Case	$O(V + E)$
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Space Complexity:

Space Complexity = $O(V)$

2. DFS — Depth First Search:

Depth First Search (DFS) is a graph traversal algorithm that explores a vertex as deeply as possible before backtracking. It can be implemented using recursion or a stack.

ALGORITHM:

1. Start from the given vertex.
2. Mark the vertex as visited.
3. Display the vertex.
4. Check its adjacent vertices.
5. If an adjacent vertex is unvisited, recursively visit it.
6. Continue until there are no more unvisited adjacent vertices.
7. Backtrack to the previous vertex.
8. Repeat until all reachable vertices are visited.

Time Complexity:

Case	Time Complexity
Best Case	$O(V + E)$
Average Case	$O(V + E)$
Worst Case	$O(V + E)$

Space Complexity:

Space Complexity = $O(V)$

OUTPUT:

```
Enter number of vertices: 4
Enter number of edges: 4
Enter edges (u v):
0 1
1 2
2 3
0 3
Enter starting vertex: 0
DFS Traversal: 0 1 2 3
BFS Traversal: 0 1 3 2

...Program finished with exit code 0
Press ENTER to exit console.
```

CONCLUSION:

Breadth First Search (BFS) and Depth First Search (DFS) were successfully implemented in C for graph traversal. BFS uses a queue and visits the vertices level by level, while DFS uses recursion or a stack to explore the graph as deeply as possible before backtracking. The practical helped in understanding how graphs can be represented using an adjacency matrix and how the visited array prevents repeated traversal of vertices. Both algorithms successfully traversed the given graph, demonstrating the basic techniques used for graph searching and traversal.